

```
In [4]: import tensorflow
        from tensorflow import keras
        from tensorflow.keras import Sequential
        from tensorflow.keras.layers import Dense, Flatten
```

```
In [6]: (X_train,y_train),(X_test,y_test) = keras.datasets.mnist.load_data()
```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>
11490434/11490434 ————— 0s 0us/step

```
In [7]: X_test.shape
```

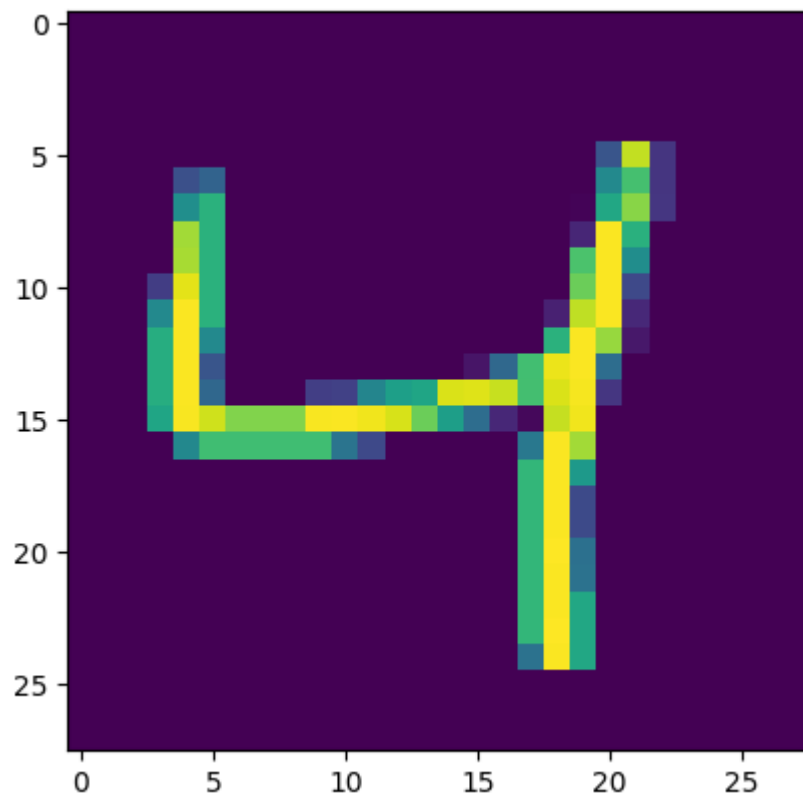
```
Out[7]: (10000, 28, 28)
```

```
In [8]: y_train
```

```
Out[8]: array([5, 0, 4, ..., 5, 6, 8], dtype=uint8)
```

```
In [9]: import matplotlib.pyplot as plt
        plt.imshow(X_train[2])
```

```
Out[9]: <matplotlib.image.AxesImage at 0x79986ac0ed10>
```



```
In [10]: X_train = X_train/255  
X_test = X_test/255
```

```
In [ ]: X_train[0]
```

```

Out[ ]: array([[0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , ],
[0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , ],
[0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , ],
[0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , ],
[0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , ],
[0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.01176471, 0.07058824, 0.07058824,
0.07058824, 0.49411765, 0.53333333, 0.68627451, 0.10196078,
0.65098039, 1.      , 0.96862745, 0.49803922, 0.      ,
0.      , 0.      , 0.      , ],
[0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.11764706, 0.14117647,
0.36862745, 0.60392157, 0.66666667, 0.99215686, 0.99215686,
0.99215686, 0.99215686, 0.99215686, 0.88235294, 0.6745098 ,

```

0.99215686, 0.94901961, 0.76470588, 0.25098039, 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.19215686, 0.93333333, 0.99215686,
0.99215686, 0.99215686, 0.99215686, 0.99215686, 0.99215686,
0.99215686, 0.99215686, 0.98431373, 0.36470588, 0.32156863,
0.32156863, 0.21960784, 0.15294118, 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.07058824, 0.85882353, 0.99215686,
0.99215686, 0.99215686, 0.99215686, 0.99215686, 0.77647059,
0.71372549, 0.96862745, 0.94509804, 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0.31372549, 0.61176471,
0.41960784, 0.99215686, 0.99215686, 0.80392157, 0.04313725,
0. , 0.16862745, 0.60392157, 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0.05490196,
0.00392157, 0.60392157, 0.99215686, 0.35294118, 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0.54509804, 0.99215686, 0.74509804, 0.00784314,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0.04313725, 0.74509804, 0.99215686, 0.2745098 ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0.1372549 , 0.94509804, 0.88235294,

0.62745098, 0.42352941, 0.00392157, 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0.31764706, 0.94117647,
0.99215686, 0.99215686, 0.46666667, 0.09803922, 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0.17647059,
0.72941176, 0.99215686, 0.99215686, 0.58823529, 0.10588235,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0.0627451 , 0.36470588, 0.98823529, 0.99215686, 0.73333333,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.97647059, 0.99215686, 0.97647059,
0.25098039, 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0.18039216,
0.50980392, 0.71764706, 0.99215686, 0.99215686, 0.81176471,
0.00784314, 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.15294118, 0.58039216, 0.89803922,
0.99215686, 0.99215686, 0.99215686, 0.98039216, 0.71372549,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,

0.09411765, 0.44705882, 0.86666667, 0.99215686, 0.99215686,
0.99215686, 0.99215686, 0.78823529, 0.30588235, 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0.09019608, 0.25882353,
0.83529412, 0.99215686, 0.99215686, 0.99215686, 0.99215686,
0.77647059, 0.31764706, 0.00784314, 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0.07058824, 0.67058824, 0.85882353, 0.99215686,
0.99215686, 0.99215686, 0.99215686, 0.76470588, 0.31372549,
0.03529412, 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0.21568627,
0.6745098 , 0.88627451, 0.99215686, 0.99215686, 0.99215686,
0.99215686, 0.95686275, 0.52156863, 0.04313725, 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0.53333333,
0.99215686, 0.99215686, 0.99215686, 0.83137255, 0.52941176,
0.51764706, 0.0627451 , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0. , 0. , 0. ,
0. , 0. , 0.],
[0. , 0. , 0. , 0. , 0. ,

```

0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      , 0.      , 0.      ,
0.      , 0.      , 0.      ]]

```

```

In [ ]: model = Sequential()

model.add(Flatten(input_shape=(28,28)))
model.add(Dense(128,activation='relu'))
model.add(Dense(32,activation='relu'))
model.add(Dense(10,activation='softmax'))

```

```

In [ ]: model.summary()

```

Model: "sequential_5"

Layer (type)	Output Shape	Param #
=====		
flatten_4 (Flatten)	(None, 784)	0
dense_11 (Dense)	(None, 128)	100480
dense_12 (Dense)	(None, 32)	4128
dense_13 (Dense)	(None, 10)	330
=====		
Total params: 104,938		
Trainable params: 104,938		
Non-trainable params: 0		

```

In [ ]: model.compile(loss='sparse_categorical_crossentropy',optimizer='Adam',metrics=['accuracy'])

```

```

In [ ]: history = model.fit(X_train,y_train,epochs=25,validation_split=0.2)

```

Epoch 1/25
1500/1500 [=====] - 5s 3ms/step - loss: 0.2845 - accuracy: 0.9168 - val_loss: 0.1425 - val_accuracy: 0.9570

Epoch 2/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.1164 - accuracy: 0.9638 - val_loss: 0.1258 - val_accuracy: 0.9605

Epoch 3/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0800 - accuracy: 0.9754 - val_loss: 0.0912 - val_accuracy: 0.9728

Epoch 4/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0594 - accuracy: 0.9817 - val_loss: 0.1014 - val_accuracy: 0.9697

Epoch 5/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0462 - accuracy: 0.9851 - val_loss: 0.1024 - val_accuracy: 0.9687

Epoch 6/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0377 - accuracy: 0.9881 - val_loss: 0.1060 - val_accuracy: 0.9712

Epoch 7/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0313 - accuracy: 0.9898 - val_loss: 0.0960 - val_accuracy: 0.9758

Epoch 8/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0283 - accuracy: 0.9902 - val_loss: 0.1061 - val_accuracy: 0.9732

Epoch 9/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0230 - accuracy: 0.9923 - val_loss: 0.1160 - val_accuracy: 0.9713

Epoch 10/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0179 - accuracy: 0.9939 - val_loss: 0.1134 - val_accuracy: 0.9761

Epoch 11/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0177 - accuracy: 0.9937 - val_loss: 0.1204 - val_accuracy: 0.9742

Epoch 12/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0156 - accuracy: 0.9947 - val_loss: 0.1103 - val_accuracy: 0.9761

Epoch 13/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0156 - accuracy: 0.9949 - val_loss: 0.1221 - val_accuracy: 0.9743

Epoch 14/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0121 - accuracy: 0.9960 - val_loss: 0.1246 - val_accuracy:


```
0.9764
Epoch 15/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0122 - accuracy: 0.9958 - val_loss: 0.1362 - val_accuracy:
0.9753
Epoch 16/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0100 - accuracy: 0.9964 - val_loss: 0.1340 - val_accuracy:
0.9772
Epoch 17/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0131 - accuracy: 0.9955 - val_loss: 0.1371 - val_accuracy:
0.9752
Epoch 18/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0101 - accuracy: 0.9968 - val_loss: 0.1318 - val_accuracy:
0.9760
Epoch 19/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0116 - accuracy: 0.9962 - val_loss: 0.1577 - val_accuracy:
0.9732
Epoch 20/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0093 - accuracy: 0.9969 - val_loss: 0.1570 - val_accuracy:
0.9718
Epoch 21/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0095 - accuracy: 0.9970 - val_loss: 0.1347 - val_accuracy:
0.9772
Epoch 22/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0076 - accuracy: 0.9975 - val_loss: 0.1591 - val_accuracy:
0.9762
Epoch 23/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0079 - accuracy: 0.9975 - val_loss: 0.1919 - val_accuracy:
0.9682
Epoch 24/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0078 - accuracy: 0.9974 - val_loss: 0.1692 - val_accuracy:
0.9737
Epoch 25/25
1500/1500 [=====] - 4s 3ms/step - loss: 0.0084 - accuracy: 0.9971 - val_loss: 0.1647 - val_accuracy:
0.9753
```

```
In [ ]: y_prob = model.predict(X_test)
```

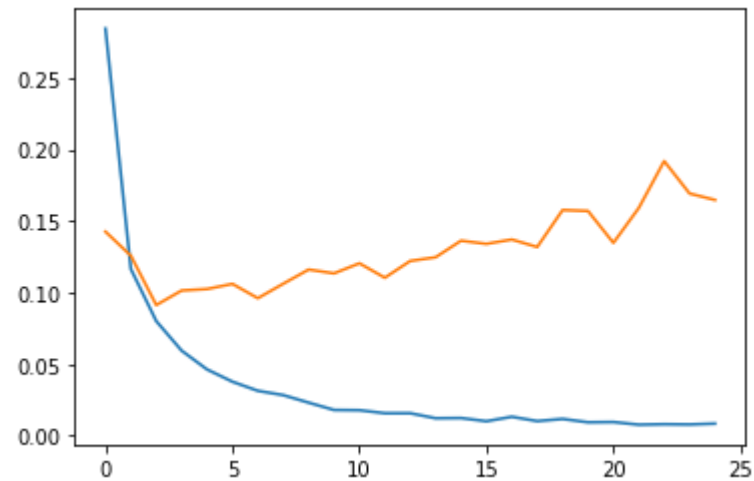
```
In [ ]: y_pred = y_prob.argmax(axis=1)
```

```
In [ ]: from sklearn.metrics import accuracy_score
accuracy_score(y_test,y_pred)
```

```
Out[ ]: 0.9755
```

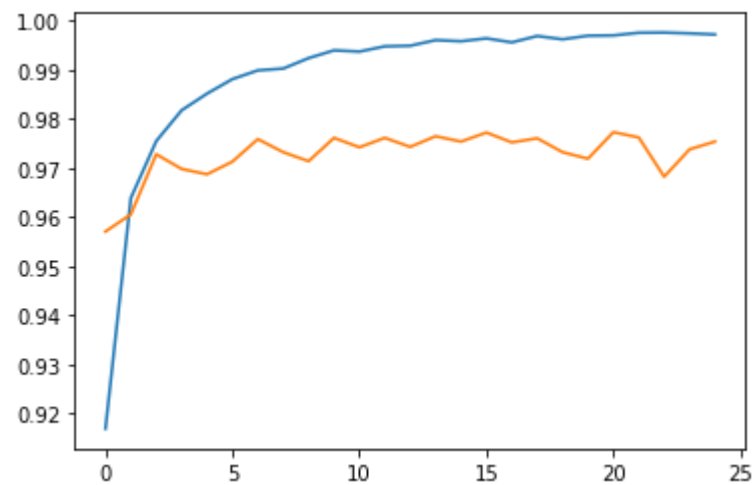
```
In [ ]: plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
```

```
Out[ ]: [<matplotlib.lines.Line2D at 0x7f5737a6c590>]
```



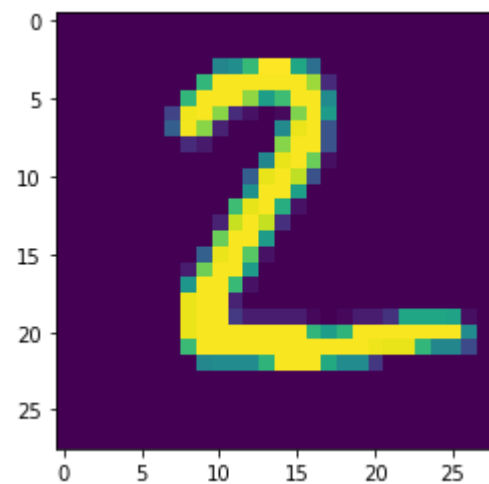
```
In [ ]: plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
```

```
Out[ ]: [<matplotlib.lines.Line2D at 0x7f5737831e50>]
```



```
In [ ]: plt.imshow(X_test[1])
```

```
Out[ ]: <matplotlib.image.AxesImage at 0x7f5736844c50>
```



```
In [ ]: model.predict(X_test[1].reshape(1,28,28)).argmax(axis=1)
```

```
Out[ ]: array([2])
```

```
In [ ]:
```

